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Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

NEET(UG)

MINOR

04-03-2023

ENTHUSIAST ADVANCE COURSE PHASE : MEA-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 Hours 20 Minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

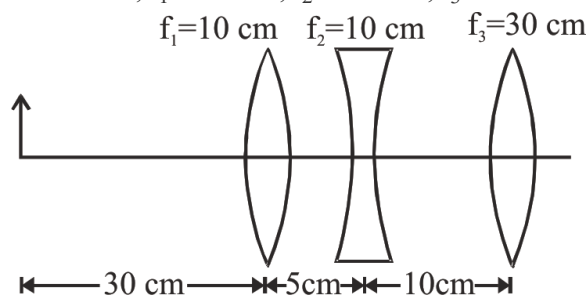
Topic : ELECTROSTATICS, RAY OPTICS

SECTION - A

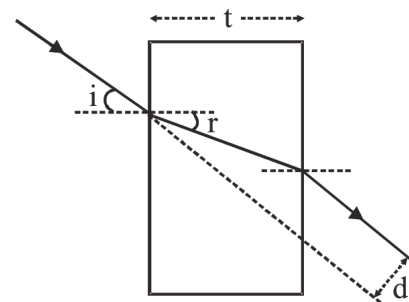
Attempt All 35 questions

- An air bubble in a glass slab ($\mu = 1.5$) is 6 cm deep as viewed from one face and 4 cm deep as viewed from the other face. The thickness of the glass slab is
 - 6.67 cm
 - 10 cm
 - 15 cm
 - Data is incomplete
- The radius of curvature of a convex spherical mirror is 1.2 m. How far away from the mirror is an object of height 1.2 cm be placed if the distance between its virtual image and the mirror is 0.3 m ?
 - 0.3 m
 - 0.5 m
 - 0.6 m
 - 0.9 m
- A *biconvex* lens has a focal length $\frac{2}{3}$ times the radius of curvature of either surface. The refractive index of material is :-
 - 1.75
 - 1.33
 - 1.5
 - 1.0
- A lens of power + 2 dipotres is placed in contact with a lens of power – 1 diopetre. The combination will behave like :-
 - A convergent lens of focal length 50 cm
 - A divergent lens of focal length 100 cm
 - A convergent lens of focal length 100 cm
 - A convergent lens of focal length 200 cm

- The position of final image formed by the given lens combination from the third lens will be at a distance of, $f_1 = +10\text{cm}$, $f_2 = -10\text{ cm}$, $f_3 = +30\text{cm}$.



- 15 cm
 - infinity
 - 45 cm
 - 30 cm
- If the refractive index of a material of equilateral prism is $\sqrt{3}$, then angle of minimum deviation of the prism is :-
 - 30°
 - 45°
 - 60°
 - 75°
 - A ray of light is incident on a thick slab of glass of thickness t as shown in the figure. The emergent ray is parallel to the incident ray but displaced sideways by a distance d . If the angles are small then d is. :-



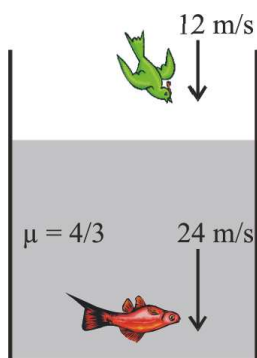
- $t \left(1 - \frac{i}{r} \right)$
- $rt \left(1 - \frac{i}{r} \right)$
- $it \left(1 - \frac{r}{i} \right)$
- $t \left(1 - \frac{r}{i} \right)$

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8. Two identical glass ($\mu_g = 3/2$) equiconvex lenses of focal length 'f' are kept in contact. The space between the two lenses is filled with water ($\mu_w = \frac{4}{3}$). The focal length of the combination is

- (1) f (2) $f/2$
(3) $4f/3$ (4) $3f/4$

9. A fish and a bird are moving as shown in figure. Find the velocity of image of bird as observed by fish.



- (1) 8 m/s upwards
(2) 8 m/s downwards
(3) 9 m/s upwards
(4) 9 m/s downwards
10. A microscope is focussed on a mark on a piece of paper and then a slab of glass of thickness 9 cm and refractive index 1.5 is placed over the mark. How should the microscope be moved to get the mark in focus again :-
- (1) 3 cm upward
(2) 4.5 cm downward
(3) 1 cm downward
(4) 2 cm upward

11. An optical fibre ($\mu = 2$) is surrounded by a glass coating ($\mu = \sqrt{3}$). Find the critical angle for total internal reflection at the fibre-glass interface.

- (1) 60° (2) 30° (3) 45° (4) 90°

12. The dispersive powers of the materials of the two lenses are in the ratio 2:3. If the achromatic combination of these two lenses in contact is a convex lens of focal length 120 cm, then the focal length of the component lenses are.

- (1) 40 cm & -60 cm
(2) -40 cm & 60 cm
(3) 20 cm & -30 cm
(4) -20 cm & 30 cm

13. **Assertion :** If the rays are diverging after emerging from a lens; the lens must be concave.

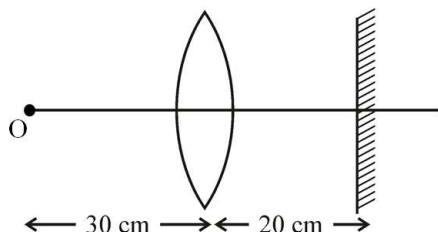
Reason : The convex lens can give only converging rays.

- (1) Both Assertion & Reason are correct & the reason is a correct explanation of the Assertion
(2) Both Assertion & Reason are correct but Reason is not correct explanation of the Assertion.
(3) Only Assertion is correct
(4) Both are incorrect

14. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a :-

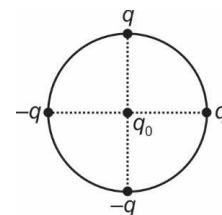
- (1) Convergent lens of focal length 3.5 R
(2) Convergent lens of focal length 3.0 R
(3) Divergent lens of focal length 3.5 R
(4) Divergent lens of focal length 3.0 R

15. A point object is placed at a distance of 30 cm from convex lens of focal length 15 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 20 cm from it, the final image would be formed at a distance of :-

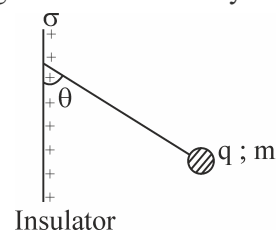


- (1) 20 cm from the lens, it would be real image
 (2) 10 cm from the lens, it would be real image
 (3) 20 cm from the plane mirror, it would be virtual image
 (4) 10 cm from the plane mirror, it would be virtual image.
16. Two convex lenses of powers 4 D and 6 D are separated by a distance of $\frac{1}{6}$ m. The power of the optical system so formed is :-
- (1) -6 D (2) +6 D
 (3) 10 D (4) 2 D
17. A convex lens is placed between object and a screen to obtain sharp image. The size of object is 3 cm and an image of height 9 cm is obtained on the screen. When the lens is displaced to a new position, what will be the size of sharp image on the screen ?
- (1) 2 cm (2) 6 cm
 (3) 4 cm (4) 1 cm

18. Four charges $q, q, -q, -q$ are placed on circumference of a circle of radius R as shown in the figure. What is electrostatic force on q_0 placed at the centre of the circle?



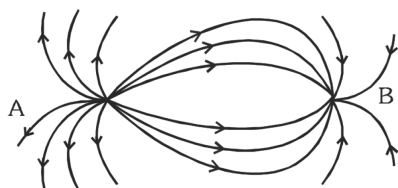
- (1) $\frac{qq_0}{2\sqrt{2}R^2\epsilon_0}$ (2) $\frac{\sqrt{8}qq_0}{4\pi\epsilon_0 R^2}$
 (3) $\frac{2qq_0}{4\pi\epsilon_0 R^2}$ (4) $\frac{qq_0}{2\sqrt{2}\epsilon_0 R^2}$
19. Find the angle θ in the shown system :



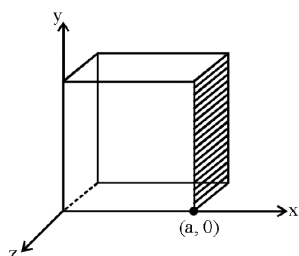
- (1) $\tan^{-1}\left(\frac{\sigma}{2\epsilon_0 mg}\right)$ (2) $\tan^{-1}\left(\frac{\sigma q}{\epsilon_0 mg}\right)$
 (3) $\tan^{-1}\left(\frac{\sigma q}{2\epsilon_0 mg}\right)$ (4) $\tan^{-1}\left(\frac{\sigma}{\epsilon_0 mg}\right)$
20. A ring of radius r carries a charge Q uniformly distributed over its length. A charge q is placed at its centre will experience a force equal to:-
- (1) $\frac{qQ}{4\pi\epsilon_0 r^2}$ (2) $\frac{qQ}{8\pi\epsilon_0 r^3}$
 (3) zero (4) None of these
21. The electric intensity outside a charged sphere of radius R at a distance r ($r > R$) is

- (1) $\frac{\sigma R^2}{\epsilon_0 r^2}$ (2) $\frac{\sigma r^2}{\epsilon_0 R^2}$ (3) $\frac{\sigma r}{\epsilon_0 R}$ (4) $\frac{\sigma R}{\epsilon_0 r}$

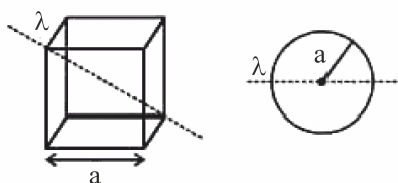
22. The spatial distribution of the electric field lines due to charges (A, B) is shown in figure. Which one of the following statements is correct ?



- (1) A is +ve and B is -ve and $|A| > |B|$
 (2) A is -ve and B is +ve ; $|A| = |B|$
 (3) Both are +ve but $A > B$
 (4) Both are -ve but $A > B$
23. If $\vec{E} = \frac{E_0 x}{a} \hat{i}$ ($x - mt$) then flux through the shaded area is :-



- (1) $E_0 a^2$ (2) Zero
 (3) $E_0 a^3$ (4) $-E_0 a^3$
24. A linear charge having linear charge density λ , penetrates a cube diagonally so as to have maximum flux and then it penetrates a sphere diametrically as shown. What will be the ratio of flux coming out of cube and sphere ?

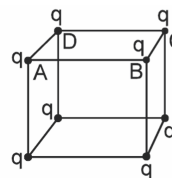


- (1) $\frac{1}{2}$ (2) $\frac{2}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{1}$

25. The electric field at a distance $\frac{3R}{2}$ from the center of a charged conducting spherical shell of radius R is E. The electric field at a distance R/2 from the center of the sphere is :

- (1) $\frac{9E}{8}$
 (2) Zero
 (3) $\frac{8E}{4}$
 (4) $\frac{E}{3}$

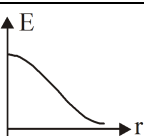
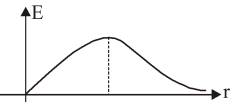
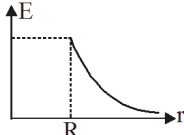
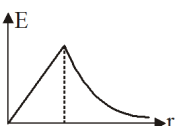
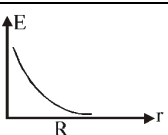
26. Eight point charges (can be assumed as uniformly charged small spheres and their centres at the corner of the cube) having value q each are fixed at vertices of a cube. The electric flux through square surface ABCD of the cube is



- (1) $\frac{q}{24\epsilon_0}$ (2) $\frac{q}{12\epsilon_0}$
 (3) $\frac{q}{6\epsilon_0}$ (4) $\frac{q}{8\epsilon_0}$
27. Consider the following statements :
- (A) If electric field is zero at a point then the potential at that point must be zero.
 (B) If electric potential is zero at a point then electric field at that point must be zero.
- (1) (A) is correct, (B) is false.
 (2) (A) is false, (B) is correct
 (3) Both (A) and (B) are wrong.
 (4) Both (A) and (B) are correct.

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28. Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in Column I.

Column-I		Column-II	
(A)	Ring along its axis	(P)	
(B)	Uniformly charged solid sphere	(Q)	
(C)	Uniformly charged spherical shell	(R)	
(D)	Combination of charge +Q and -Q at the perpendicular bisector	(S)	
		(T)	

- (1) A→P, B→Q, C→R, D→S
 (2) A→S, B→Q, C→P, D→R
 (3) A→Q, B→S, C→R, D→T
 (4) A→Q, B→S, C→R, D→P
29. When the separation between two charges is increased, the electric potential energy of the charges:-
 (1) Increases
 (2) Decreases
 (3) Remain the same
 (4) May increase or decrease
30. Three charges $2q$, $-q$, $-q$ are located at the vertices of an equilateral triangle. At the centre of the triangle
 (1) The field is zero but potential is non-zero
 (2) The field is non-zero but potential is zero
 (3) Both field and potential are zero
 (4) Both field and potential are non-zero

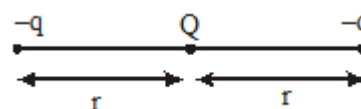
31. Two charges $-6q$ and $+2q$ are placed at the points P (2, 0) cm. and Q (6, 0) cm. on X-axis, then potential at the point A (9, 0) cm. is –

- (1) Positive (2) Negative
 (3) Zero (4) None of the above

32. Two spheres of radii 2 cm and 3 cm are charged to same potential. If σ_1 and σ_2 be respectively the value of surface charge density on the conductors, then the ratio $\frac{\sigma_1}{\sigma_2}$ will be :-

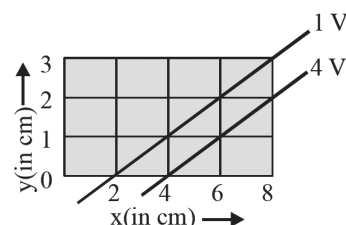
- (1) $\frac{4}{9}$ (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) $\frac{9}{4}$

33. Three charges are placed as shown in fig if the electric potential energy of system is zero, then Q : q-



- (1) $\frac{Q}{q} = \frac{-2}{1}$ (2) $\frac{Q}{q} = \frac{2}{1}$
 (3) $\frac{Q}{q} = \frac{-1}{2}$ (4) $\frac{Q}{q} = \frac{1}{4}$

34. Figure shows two equipotential lines in xy-plane for an electric field. The x-component E_x and y-component E_y of the field in the electric space between these equipotential lines are, respectively-



- (1) +150 V/m, -300 V/m
 (2) -150 V/m, +300 V/m
 (3) +200 V/m, 100 V/m
 (4) -200 V/m, -100 V/m

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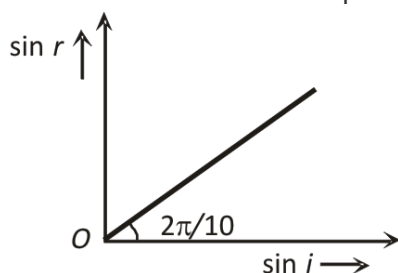
35. 8 charges of equal magnitude 'q' are placed at the corners of a cube of side 'a'. Find work done by field in bringing the charge 'Q' from infinite to the center of the cube –

(1) $\frac{8 kQq}{\sqrt{3} a}$ (2) $\frac{16 kQq}{\sqrt{3} a}$
 (3) $\frac{-16 kQq}{\sqrt{3} a}$ (4) $\frac{-8 kQq}{\sqrt{3} a}$

SECTION - B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

36. The graph between sine of angle of refraction ($\sin r$) in medium 2 and sine of angle of incidence ($\sin i$) in medium 1 indicates that ($\tan 36^\circ \approx \frac{3}{4}$)



- (1) Total internal reflection can take place
 (2) Total internal reflection cannot take place
 (3) Any of (1) and (2)
 (4) Data is incomplete
37. A tank is filled with benzene to a height of 120 mm. The apparent depth of a needle lying at a bottom of the tank is measured by a microscope to be 80 mm. The refractive index of benzene is
- (1) 1.5 (2) 2.5 (3) 3.5 (4) 4.5

38. Light enters at an angle of incidence in a transparent rod of refractive index n . For what value of the refractive index of the material of the rod the light once entered into it will not leave it through its lateral face whatsoever be the value of angle of incidence.

(1) $n > \sqrt{2}$ (2) $n = 1$
 (3) $n = 1.1$ (4) $n = 1.3$

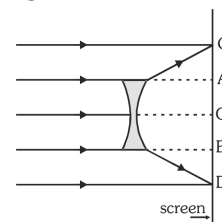
39. A slab of glass, thickness 6 cm and refractive index 1.5, is placed in front of a concave mirror, the faces of the slab being perpendicular to the principal axis of the mirror. If the radius of curvature of the mirror is 40 cm and the reflected image coincides with the object, then the distance of the object from the mirror is :-

(1) 30 cm (2) 22 cm (3) 42 cm (4) 28 cm

40. Two concave lenses are kept in contact with each other. If the space between the two lenses is filled with a material of smaller refractive index, the magnitude of the focal length of the combination

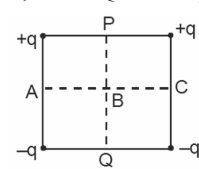
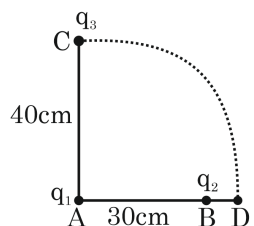
(1) Becomes undefined (2) remains unchanged
 (3) increases (4) decreases

41. A concave lens is placed in the path of a uniform parallel beam of light falling on a screen as shown. Then,



- (1) intensity of light will be the same everywhere on the screen.
 (2) intensity in region AB will be smaller than what it would be in the absence of the lens
 (3) in the region AC and BD, the intensity will be smaller than what it would be in the absence of the lens
 (4) None of these

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42. A screen is placed at a distance 40 cm away from an illuminated object. A converging lens is placed between the source and the screen and it is attempted to form the image of the source on the screen. If no position could be found, the focal length of the lens
- Must be less than 10 cm
 - Must be greater than 10 cm
 - Must not be greater than 20 cm
 - Must not be less than 10 cm
43. A point source 's' is placed at a distance of 15 cm from a converging lens of focal length 10 cm on its principal axis. where should a diverging mirror of focal length 12 cm be placed so that a real image is formed on the source itself?
- 80 cm from lens
 - 6 cm from lens
 - 24 cm from lens
 - None
44. Two objects are rubbed against each other, the nature of electric force, when they are placed at some distance is :
- Attractive
 - Repulsive
 - Both (1) and (2)
 - Either (1) or (2)
45. Two point charges repel each other with a force of 100 N. One of the charges is increased by 10%, and the other is reduced by 10%. The new force of repulsion at the same distance would be :
- 100 N
 - 121 N
 - 99 N
 - None of these
46. A drop of water of mass 10^{-6} Kg has an electric charge of 1000nC. What electric field will balance the drop in air –
- 10 V/m upwards
 - 10 V/m downwards
 - 0.1 V/m downwards
 - 0.1 V/m upwards
47. Eight charges, $1\mu\text{C}$, $-7\mu\text{C}$, $-4\mu\text{C}$, $10\mu\text{C}$, $2\mu\text{C}$, $-5\mu\text{C}$, $-3\mu\text{C}$ and $6\mu\text{C}$ are situated at the eight corners of a cube of side 20 cm. A spherical surface of radius 80 cm encloses this cube. The centre of the sphere coincides with the centre of the cube. Then the total outgoing flux from the spherical surface (in unit of volt meter) is :
- $36\pi \times 10^3$
 - $684\pi \times 10^3$
 - zero
 - none of the above
48. Two infinite linear charges are placed parallel at 0.1 m apart. If each has charge density of $5\mu\text{ C/m}$, then the force per unit length of one of linear charges in N/m is :
- 2.5
 - 3.25
 - 4.5
 - 7.5
49. Figure represents a square carrying charges $+q$, $+q$, $-q$, $-q$ at its four corners as shown. Then the potential will be zero at points : (A, C, P and Q are mid points of sides)
- 
- A, B, C, P and Q
 - A, B and C
 - A, P, C and Q
 - P, B and Q
50. Two charges q_1 and q_2 are placed 30 cm apart, as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of the system is $\frac{q_3}{4\pi\epsilon_0} k$, where k is:
- 
- $8q_1$
 - $6q_1$
 - $8q_2$
 - $6q_2$

Topic : SOLUTION, PERIODIC TABLE (APPLICATION OF EN), CHEMICAL BONDING (UPTO d-ORBITAL, CONTRACTION, BENT RULE)

SECTION-A

Attempt All 35 questions

51. If inter nuclear axis is x-axis, which of the following overlapping may result in formation of O_2 molecule :-

- (A) $1s-1s$ (B) $2p_x-2p_x$ (C) $2p_y-2p_y$
(D) $3p_z-3p_z$ (E) $3d_{xy}-3d_{xy}$

- (1) B, C, D (2) B, C, D, E
(3) B, C (4) B, E

52. $O \overset{(I)}{\text{---}} O \overset{(II)}{\text{---}} O$

In above molecule of ozone (O_3), the bond order of (I) and (II) are :-

- (I) (II)
(1) > 2 > 2
(2) < 2 < 2
(3) $= 2$ $= 2$
(4) > 2 < 2

53. Which of the following molecule/ion is planar :

- (1) PO_4^{3-} (2) H_3O^+ (3) XeF_3^+ (4) $XeOF_4$

54. Given below two statements :

Statement-I: The formation of covalent bond between two atoms results by transfer of electrons.

Statement-II: In general, greater the overlap the stronger is the bond formed between two atoms.

In light of the above statements choose the most appropriate answer from the option given below.

- (1) Both statement-I and statement-II are correct
(2) Both statement-I and statement-II are incorrect
(3) Statement-I is correct and statement-II is incorrect
(4) Statement-I is incorrect and statement-II is correct

55. Hypovalent species among the following is :

- (1) $SnCl_2$ (2) $SnCl_4$
(3) $CuCl$ (4) BH_4^-

56. Which of the following statements is incorrect ?

- (1) ClF_3 has T-shape
(2) In SF_4 , F-S-F equatorial bond angle is less than 120°
(3) In $[ICl_4]^-$, Cl-I-Cl bond angle is 109°
(4) Shape of I_3^- molecule is linear

57. The number of 90° angle in cationic part of solid PCl_5 is :-

- (1) 10 (2) 12 (3) 8 (4) Zero

58. Match the column :

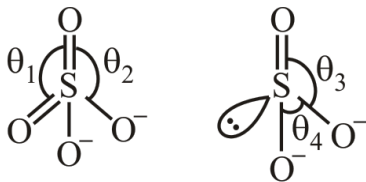
Compound		Properties	
(A)	N_2O	(P)	sp Hybridisation
(B)	$Sn(OH)_4$	(Q)	Neutral
(C)	NO_2	(R)	Amphoteric
(D)	ClO_2	(S)	Acidic

- (1) $A \rightarrow P, Q; B \rightarrow R; C \rightarrow S; D \rightarrow S$
(2) $A \rightarrow P, Q; B \rightarrow P, R; C \rightarrow S; D \rightarrow R, S$
(3) $A \rightarrow P; B \rightarrow R; C \rightarrow S; D \rightarrow Q$
(4) $A \rightarrow P, Q, R; B \rightarrow R; C \rightarrow P, S; D \rightarrow R$

59. Which of the following compounds is polar as well as planar?

- (1) XeF_4 (2) SF_4 (3) H_2S (4) CH_2F_2

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60. A compound having three surrounding atom can never form which shape :-
 (1) Trigonal pyramidal
 (2) Trigonal bipyramidal
 (3) Trigonal planar
 (4) T-shape
61. Amongst the following, how many species have unequal bond lengths?
 ClF_3 , SOF_2 , SiF_4 , SF_4 , PCl_3 , PCl_5 , CO_3^{2-} , IF_7
 (1) 5 (2) 4 (3) 8 (4) 6
62. NH_3 is Isostructural with :-
 (A) SeO_3^{2-} (B) ClF_3 (C) CH_3^+ (D) CCl_3^-
 (1) Only B (2) Only A and D
 (3) Only A, B and D (4) Only D
63. Which of the following triatomic molecule has bond angle greater than $109^\circ 28'$?
 (1) C_2H_2 (2) BF_3
 (3) PH_3 (4) CO_2
64. Species with non zero dipole moment is :-
 (1) PCl_3F_2 (2) PCl_5
 (3) PCl_4^+ (4) PF_3Cl_2
65. Central atom is present in its highest excited state in :-
 (1) XeF_4 (2) SF_4
 (3) SO_3 (4) IF_5
66. Octet rule is followed by :-
 (1) XeF_2 (2) NO_3^-
 (3) SF_4 (4) PF_5
67. Select the correct option regarding given species :-

 (1) $\theta_1 > \theta_2$ (2) $\theta_2 > \theta_3$
 (3) $\theta_2 = \theta_3$ (4) $\theta_3 > \theta_4$
68. Vapour pressure of pure benzene is 800 mm Hg. If 3g of non-volatile solute is added in 19.5 g of benzene, the vapour pressure is reduced to 760 mm Hg. Calculate molar mass of solute ?
 (1) 128 g (2) 300 g
 (3) 400 g (4) 228 g
69. Osmotic pressure of 0.4 M CH_3COOH solution is found to be 15 atm at 300K. Degree of dissociation of CH_3COOH is :
 (1) 0.25 (2) 0.50 (3) 0.75 (4) 0.9
70. 10% solution of urea is isotonic with 6% solution of a non-volatile solute X, what is the molecular mass of solute X?
 (1) 6 g mol^{-1} (2) 60 g mol^{-1}
 (3) 36 g mol^{-1} (4) 32 g mol^{-1}
71. Mole fraction of a solute in an aqueous solution is 0.2. The molality of the solution will be
 (1) 13.88 (2) 1.388
 (3) 0.138 (4) 0.0138
72. A gas X above water has partial pressure 2 atm value of Henry's constant $K_H = 5 \text{ atm}$ then find out mole fraction of gas X in water?
 (1) 0.5 (2) 0.4 (3) 0.2 (4) 0.1

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73. If the total vapour pressure of the liquid mixture A and B is given by the equation: $P = 180X_A + 90 \text{ mm Hg}$ then the ratio of the vapour pressure of the pure liquids A and B is given by
 (1) 3 : 2 (2) 4 : 1 (3) 3 : 1 (4) 6 : 2
74. Find depression in freezing point of 20% by weight acetic acid solution in benzene assuming complete association [K_f for benzene is 5.12 K/m]
 (1) 8.53°C (2) 5.3°C
 (3) 10.6°C (4) 1.06°C
75. Which one of the following aqueous solutions will exhibit highest boiling point?
 (1) 0.01 M KNO_3 (2) $0.01 \text{ M Na}_2\text{SO}_4$
 (3) 0.015 M urea (4) 0.015 M glucose
76. Which of the following is not an example of maximum boiling azeotropic mixture.
 (1) $\text{H}_2\text{O} + \text{HCl}$
 (2) $\text{H}_2\text{O} + \text{HNO}_3$
 (3) $\text{CHCl}_3 + \text{CH}_3\text{COCH}_3$
 (4) $\text{CH}_3\text{OH} + \text{C}_2\text{H}_5\text{OH}$
77. **Statement-I** : When methyl alcohol is added to water, boiling point of water increases.
Statement-II : When a volatile solute is added to a volatile solvent elevation in boiling point is observed.
 (1) Statement-I and statement-II both are true.
 (2) Statement-I is true and statement-II is false.
 (3) Statement-I is false and statement-II is true.
 (4) Statement-I and statement-II both are false.
78. Maximum amount of solid solute that can be dissolved in specific amount of a given liquid solvent doesn't depend upon :
 (1) Temperature
 (2) Nature of solute
 (3) Pressure
 (4) Nature of solvent
79. The value of Henry constant K_H is
 (1) Greater for gases with higher solubility.
 (2) Constant for all the gases.
 (3) Not related to solubility of gases.
 (4) Increases with increase in temperature.
80. **Statement-I** :- When NaCl is added to water a depression in freezing point is observed.
Statement-II : The lowering of vapour pressure of a solution causes depression in freezing point.
 (1) Statement-I and statement-II both are true.
 (2) Statement-I is true and statement-II is false.
 (3) Statement-I is false and statement-II is true.
 (4) Statement-I and statement-II both are false.
81. Osmotic pressure of a solution prepared by dissolving 0.1 mole of K_2SO_4 in 2.5 L of water at 27°C , assuming that it is completely dissociated.
 (1) 3 atm (2) 25 atm (3) 30 atm (4) 50 atm
82. Which of the following show minimum freezing point?
 (1) 1% glucose in water
 (2) 1% sucrose in water
 (3) 1% urea in water
 (4) 1% NaCl in water

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83. A solution of A and B has equal moles of A and B. If vapour pressure of solution is 270 mm of Hg then which of the following is incorrect : (Given $P_A^0 = 200$ mm and $P_B^0 = 400$ mm of Hg)

- (1) $\Delta_{\text{sol}} H < 0$
 (2) $\Delta G < 0$
 (3) $\Delta S > 0$
 (4) $P_{\text{obs}} - P_{\text{calculated}} = 0$

84. What is the % (w/v) of 10'V' H_2O_2 solution ?

- (1) 30.3%
 (2) 1.5%
 (3) 3.03%
 (4) 0.89%

85. 6g of urea and 18g of glucose is dissolved to make 500 mL of solution at 300 K. Calculate osmotic pressure.

- (1) 10 atm (2) 8 atm (3) 3 atm (4) 12 atm

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

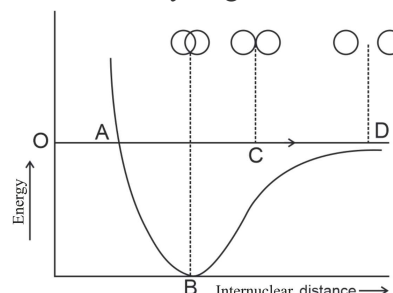
86. In which of the following conversion change in hybridisation is not observed :-

- (1) $\text{AlCl}_3 + \text{Cl}^- \rightarrow \text{AlCl}_4^-$
 (2) $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{B} \cdot \text{NH}_3$
 (3) $\text{BeF}_2 + 2\text{F}^- \rightarrow \text{BeF}_4^{2-}$
 (4) $\text{SO}_2 + (\text{O}) \rightarrow \text{SO}_3$

87. Species in which lone pair of central atom has maximum % s-character is :

- (1) NH_2^- (2) PH_3 (3) NH_3 (4) PH_4^+

88. Look at the following potential energy curve which of the following correctly represents the most stable state of hydrogen molecule.



- (1) A (2) B (3) C (4) D

89. Match the following

	Species		Properties
(A)	SO_2	(P)	Planar
(B)	XeOF_4	(Q)	Polar
(C)	N_2O	(R)	$p\pi - p\pi$ bond is present
(D)	ClF_3	(S)	d-orbital is used in hybridisation
		(T)	Lone pair is present on central atom

- (1) $A \rightarrow P, R, T$; $B \rightarrow S, T$; $C \rightarrow P, Q, R$; $D \rightarrow P, Q, S$
 (2) $A \rightarrow P, Q, R, S, T$; $B \rightarrow P, Q, T$; $C \rightarrow R, S, T$; $D \rightarrow P, Q$
 (3) $A \rightarrow P, Q, R, T$; $B \rightarrow Q, S, T$; $C \rightarrow P, Q, R$; $D \rightarrow P, Q, S, T$
 (4) $A \rightarrow R, T$; $B \rightarrow Q, T$; $C \rightarrow P, R, T$; $D \rightarrow Q, S, T$

90. Which of the following orbital of central atom does not involve in bonding in XeO_3F_2 molecule :-

- (1) d_{zx} (2) d_{yz} (3) d_{xy} (4) $d_{x^2-y^2}$

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91. The true statements from the following is/are :-
 (a) PH_5 and XeF_3^- do not exist
 (b) $p\pi - d\pi$ bond is present in SO_2
 (c) XeF_4 and CH_4 have same shape
 (d) I_3^+ has bent geometry
 (1) a, c (2) a, b, c
 (3) a, c, d (4) a, b, d
92. Select the correct order of bond angle :-
 (1) $\text{BrO}_3^- > \text{IO}_3^- > \text{ClO}_3^-$
 (2) $\text{IO}_3^- > \text{BrO}_3^- > \text{ClO}_3^-$
 (3) $\text{IO}_3^- > \text{ClO}_3^- > \text{BrO}_3^-$
 (4) $\text{ClO}_3^- > \text{BrO}_3^- > \text{IO}_3^-$
93. Incorrect order of bond angle is :-
 (1) $\text{ClO}_3 > \text{ClO}_3^-$
 (2) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
 (3) $\text{PF}_3 > \text{PCl}_3 > \text{PBr}_3$
 (4) $\text{ClO}_2 > \text{ClO}_3$
94. Arrange in increasing order for osmotic pressure
 (a) 3 M NaCl (b) 1 M K_2SO_4
 (c) 1 M Sucrose (d) 2 M Urea
 (1) $c < d < b < a$ (2) $a < c < d < b$
 (3) $a < b < c < d$ (4) $d < a < c < b$
95. The rise in the boiling point of a solution containing 1.8 gram of glucose in 100 g of a solvent is 0.1°C . The molal elevation constant of the liquid is
 (1) 0.01 K/m (2) 0.1 K/m
 (3) 1 K/m (4) 10 K/m
96. Calculate the molarity of 10% by mass NaOH solution of density 2 g/ml –
 (1) 2.5 M (2) 5 M (3) 1 M (4) 0.5 M
97. Two liquids A and B have P_A° and P_B° in the ratio of 1 : 3 and the ratio of number of moles of A and B in liquid phase are 1 : 3 then mole fraction of 'A' in vapour phase in equilibrium with the solution is equal to :
 (1) 0.1 (2) 0.2 (3) 0.5 (4) 1.0
98. 500 ml of 3M CH_3COOH solution is mixed with 500 ml 12% $\frac{w}{v}$ CH_3COOH solution then calculate the final molarity of solution.
 (1) 2.5 M (2) 2.2 M
 (3) 3 M (4) 1.01 M
99. The value of Henry's law constant for some gases at 293 K is given below. Arrange the gases in the increasing order of their solubility.
 He : 144.97 kbar, H_2 : 69.16 kbar, N_2 : 76.48 kbar, O_2 : 34.86 kbar
 (1) $\text{He} < \text{N}_2 < \text{H}_2 < \text{O}_2$
 (2) $\text{O}_2 < \text{H}_2 < \text{N}_2 < \text{He}$
 (3) $\text{H}_2 < \text{N}_2 < \text{O}_2 < \text{He}$
 (4) $\text{He} < \text{O}_2 < \text{N}_2 < \text{H}_2$
100. An ideal solution has two components A and B. If A is more volatile than B and also $P_A^\circ > P_B^\circ$, then the correct relation between mole fraction of A in liquid (X) and vapour (Y) phase is :-
 (1) $X_A = Y_A$
 (2) $X_A > Y_A$
 (3) $X_A < Y_A$
 (4) Nothing can be said

Topic : HUMAN REPRODUCTION, REPRODUCTIVE HEALTH, ORIGIN AND EVOLUTION, HUMAN HEALTH AND DISEASES, SEXUAL REPRODUCTION IN FLOWERING PLANTS, REPRODUCTION IN ORGANISM

SECTION-A

Attempt All 35 questions

101. Primary and secondary immune responses are carried out with the help of _____?
- (1) T-lymphocytes (2) B-Lymphocytes
(3) 1 and 2 both (4) Natural killer cells
102. Which antibody is present in colostrum ?
- (1) Ig G (2) Ig A
(3) Ig D (4) Ig A & Ig G
103. Which cells are considerably decreased in AIDS patient ?
- (1) Macrophages (2) Helper T-cells
(3) CD₈-Cells (4) NK-Cells
104. Which of the following cells act as a HIV factory ?
- (1) CD₄-Cells (2) Macrophages
(3) CD₈-Cells (4) Neutrophils
105. Which of the following is not an example of trisomy ?
- (1) Down's syndrome (2) Turner's syndrome
(3) Klinefelter syndrome (4) Edward syndrome
106. Which of the following is not a common symptom of typhoid ?
- (1) Weakness
(2) Sustained high fever (39–40°F)
(3) Constipation
(4) Loss of appetite
107. Genetic material of HIV is
- (1) RNA(ds) (2) DNA(ds)
(3) DNA(ss) (4) RNA(ss)
108. In malignant tumors the cells proliferate, grow rapidly and move to other parts of the body to form new tumors, this stage of disease is called as ?
- (1) Metachrosis (2) Metastasis
(3) Metamorphosis (4) Contact inhibition
109. Which of the following hallucinogen is obtained from fungi ?
- (1) Ganja (2) Atropa belladonna
(3) LSD (4) Datura
110. Smoking leads to the hypersecretion mainly of ?
- (1) Histamin (2) Thyroxine
(3) Serotonin (4) Adrenaline
111. Which of the following combination is of depressant ?
- (1) Heroin, Charas (2) Morphine, Marijuana
(3) Caffeine, Hashish (4) Opium, Heroin
112. Malignant malaria is caused by
- (1) Plasmodium Malaria
(2) Plasmodium Vivax
(3) Plasmodium Falciparum
(4) Plasmodium Ovale
113. A patient is diagnosed with internal bleeding, muscular pain, anemia, blockage of intestinal passage he is most likely suffering from ____?
- (1) Filariasis (2) Typhoid
(3) Amoebiasis (4) Ascariasis
114. Select the set of correct statements
- (A) Cancer can't be induced by chemical agents
(B) Biopsy and histopathological study are important tools for the diagnosis of cancer
(C) Metastasis is not the property of benign tumors
- (1) C,A (2) A,B (3) B,C (4) A,B,C

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115. Which of the following explanation is/are related with bone marrow and thymus ?
- (a) Immature lymphocytes differentiate into antigen sensitive lymphocytes.
- (b) Provide micro environment for the development and maturation of T-lymphocyte.
- (c) Provide the sites of interaction of lymphocyte with the antigen.
- (d) Act as a filter of the body by trapping blood borne microorganisms.

- (1) a, b, c & d (2) a & b only
- (3) a only (4) a, b & c only

116. Read the following statements :

- (A) Interferons are secreted by viral infected cell and protect non infected cell from further viral infection.
- (B) Greeks like Hippocrates disproved the good humor hypothesis of health.
- (C) Cellular immune response is responsible for graft rejection.
- (D) Antibody is represented as H_2L_2 .
- How many statement are correct ?

- (1) One (2) Two (3) Three (4) Four

117. Given below are two statements –

Statement-I : Ringworm are generally acquired from bite by the female mosquito vectors.

Statement-II : Acid in the stomach, saliva in the mouth tears from eyes all prevent microbial growth. In the light of the above statements choose the most appropriate answer from the options given below :

- (1) Both Statement-I and Statement-II are incorrect
- (2) Statement-I is correct but Statement-II is incorrect
- (3) Statement-I is incorrect but Statement-II is correct
- (4) Both Statement-I and Statement-II are correct

118. Which of the following pair is matched correctly ?

- (1) Common cold → it affects lungs
- (2) Pneumonia → it affects only throat
- (3) Malaria → Chill and high fever
- (4) Typhoid → Blockage of intestinal passage

119. Match the correct:-

(a)		(i)	Cannabis sativa
(b)		(ii)	Datura
(c)		(iii)	Opium poppy

- (1) (a) - (iii), (b) - (ii), (c) - (i)
- (2) (a) - (iii), (b) - (i), (c) - (ii)
- (3) (a) - (i), (b) - (iii), (c) - (ii)
- (4) (a) - (i), (b) - (ii), (c) - (iii)

120. Match the following columns :-

(i)	Wuchereria bancrofti	(a)	Lesion are accompanied by intense itching
(ii)	Amoebiasis	(b)	Lips and finger nails may turn gray to bluish colour
(iii)	Pneumonia	(c)	Cause slowly developing chronic inflammation
(iv)	Ringworm	(d)	Housefly act as mechanical carrier

- (1) (i) - c, (ii) - d, (iii) - a, (iv) - b
- (2) (i) - a, (ii) - b, (iii) - d, (iv) - c
- (3) (i) - c, (ii) - d, (iii) - b, (iv) - a
- (4) (i) - a, (ii) - b, (iii) - c, (iv) - d

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121. Given below are pairs of pathogens diseases and causative agent category.

Which of these is incorrectly matched ?

	Disease	Pathogen	Causative agent
1	Typhoid	Salmonella	Bacteria
2	Filariasis	Microsporium	Virus
3	Ascariasis	Ascaris	Round worm
4	Common cold	Rhinovirus	Virus

122. Match the following drugs with their source –

Drug	Source
i Cannabinoids	(A) Claviceps purpurea
ii Cocaine	(B) Poppy
iii Opioids	(C) Hemp plant
iv LSD	(D) Erythroxylum

- (1) i – (D), ii – (A), iii – (B), iv – (C)
 (2) i – (C), ii – (D), iii – (B), iv – (A)
 (3) i – (D), ii – (C), iii – (B), iv – (A)
 (4) i – (C), ii – (A), iii – (D), iv – (B)

123. Match these columns :

Column-I	Column-I
A. Acid in stomach and tears from eye	i. Cellular immunity
B. Antibody	ii. Physical barrier
C. T-lymphocyte	iii. Humoral immunity
D. Mucous Coating	iv. Physiological barrier

- (1) A-iv, B-ii, C-iii, D-i
 (2) A-ii, B-iv, C-i, D-iii
 (3) A-iv, B-iii, C-i, D-ii
 (4) A-iv, B-i, C-ii, D-iii

124. Which of the following is correct statement ?

- (1) Passive immunity is slow and takes time to give its full effective response.
 (2) Osteoarthritis which affects many people in our society is an autoimmune disease.
 (3) After maturation the lymphocytes migrate to primary lymphoid organs.
 (4) The exaggerated response of the immune system to certain antigens present in the environment is called allergy.

125. Which of the following pair is not matched correctly ?

- (1) Infective stage of plasmodium for human → Sporozoite
 (2) Infective stage of plasmodium for mosquito → Gametocyte
 (3) Causative agent of malaria → Female anopheles mosquito
 (4) House flies act as a mechanical carrier → Amoebiasis

- 126.



(A)



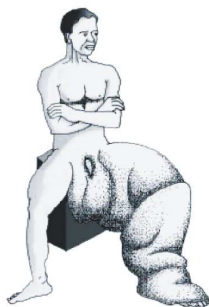
(B)

Choose the correct option about given above :-

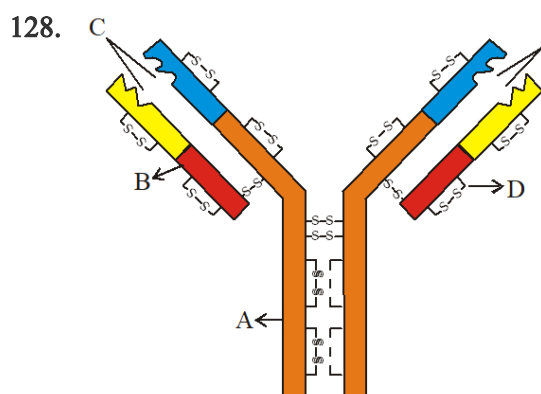
- (1) (A)-Hallucinogen (B)-Bhang
 (2) (A)-Morphine (B)-Datura
 (3) (A)-Heroin (B)- Bhang
 (4) (A)- Datura (B)-Morphine

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127. Which of the given statement is false for the disease shown in the given figure ?



- (1) It is chronic in nature
- (2) It is caused by wuchereria, the filarial worm
- (3) It is transmitted via contaminated food and water
- (4) It affects lymphatic vessels of the body.



Identify A,B,C,D in the given figure :

	A	B	C	D
(1)	Light chain	Heavy chain	Paratope	Intra chain disulphide bond
(2)	Heavy chain	Light chain	Paratope	Intra chain disulphide bond
(3)	Heavy chain	Light chain	Paratope	Intra chain disulphide bond
(4)	Heavy chain	Light chain	Epitope	Intra chain disulphide bond

129. Read the following statements (A-D)

A. The programmes called family planning were initiated in 1951
 B. In the last century an all-round development in various fields significantly improved quality of life of people
 C. Successful implementation of various action plans to attain reproductive health requires strong infrastructural facilities, professional expertise and material support.
 D. Statutory ban on amniocentesis for sex determination to legally check increasing menace of female foeticides.

Which of the following statements are correct ?

- (1) A, B, C, only
- (2) A, B,C, D,
- (3) A, B, only
- (4) B, C, D only

130. Which statement is false for amniocentesis ?

- (1) It can detect haemophilia
- (2) It can determine sex of the foetus.
- (3) It can detect all types of brain and heart disorders.
- (4) It can detect neural tube defects.

131. Which pair is not matched correctly

- (1) Population of world in 1900 $\Rightarrow$ 2000 million
- (2) Population of India in 1947 $\Rightarrow$ crossed 1 billion
- (3) Population of world in 2000 $\Rightarrow$ 6000 million
- (4) Population of world in 2011 $\Rightarrow$ 7200 million

132. The blastomeres in the blastocyst are arranged into an outer layer, termed as ?

- (1) Inner cell mass
- (2) Trophoblast
- (3) Membrana granulosa
- (4) Corona radiata

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

133. **Assertion (A)** : Estrogens and progestogens are increased several folds in maternal blood during pregnancy.

Reason (R) : They are required for the maintenance of pregnancy.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

134. **Statement I** :- After implantation finger like projections appear on inner cell mass called chorionic villi

Statement II :- In the later stage of pregnancy a hormone called hCG is also secreted by the ovary.

Choose the correct answer from the given options :-

- (1) Both Statement I and Statement II are incorrect
- (2) Both Statement I and Statement II are correct
- (3) Statement I is correct but Statement II is not correct
- (4) Statement I is incorrect but Statement II is correct

135. Which of the following statement is false ?

- (1) When morula is embedded in the endometrium of the uterus, it is termed as implantation
- (2) The inner cell mass contains certain cells called stem cells which have the potency to give rise to all the tissues and organs
- (3) The milk produced during the initial few days of lactation is called colostrum
- (4) Placenta also acts as an endocrine gland. and produces several hormones.

136. Which of the following is not a secondary lymphoid organ ?

- (1) Spleen
- (2) Lymph nodes
- (3) Thymus
- (4) Tonsils

137. Pneumonia is caused by ?

- (1) *Streptococcus Pneumoneae*
- (2) *Salmonella typhi*
- (3) *Haemophilus influenzae*
- (4) 1 and 3 both

138. Typhoid fever could be confirmed by ?

- (1) ELISA
- (2) Histopathological study
- (3) Widal test
- (4) MRI

139. Which of the following drug is used for the treatment of cancer ?

- (1) Zidovudin
- (2) Antihistaminics
- (3) Antibiotics
- (4) Taxol

140. Abrupt discontinuation of regular dose of drugs / alcohol generates.

- (1) Euphoria
- (2) Permanent feeling of well being
- (3) Hallucination
- (4) Withdrawal syndrome

141. Which of the following is infective stage of plasmodium for mosquito ?

- (1) Sporozoite
- (2) Merozoite
- (3) Gametocyte
- (4) Trophozoite

142. **Assertion:** AIDS is not a congenital disease.
Reason: AIDS is the deficiency of immune system, acquired during the lifetime of an individual.

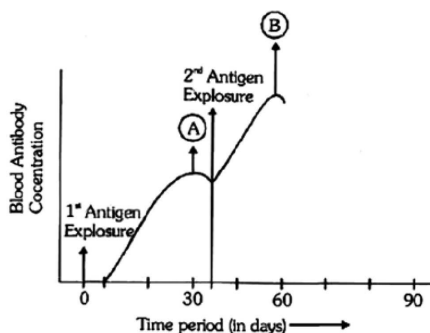
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

143. Which of the following is correct matching ?

	Column-I		Column-II
A.	Wuchereria bancrofti	i.	Neoplastic transformation
B.	Malaria	ii.	Filariasis
C.	Cancer	iii.	Contaminated needles
D.	AIDS	iv.	Plasmodium

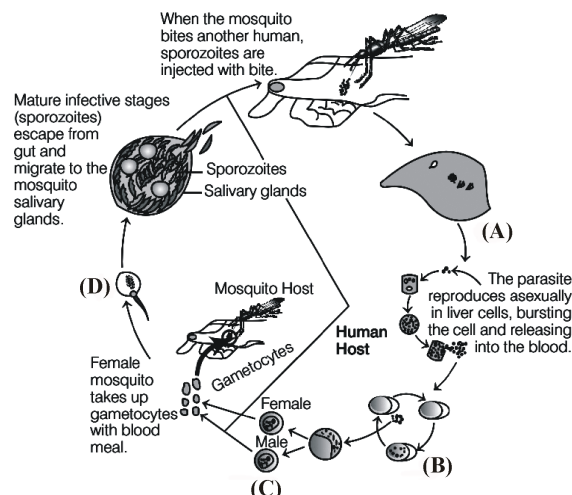
- (1) A – ii, B – iii, C – iv, D – i
- (2) A – iii, B – iv, C – i, D – ii
- (3) A – ii, B – iv, C – i, D – iii
- (4) A – ii, B – iv, C – iii, D – i

144. How many of the following are correct about the given diagram ?



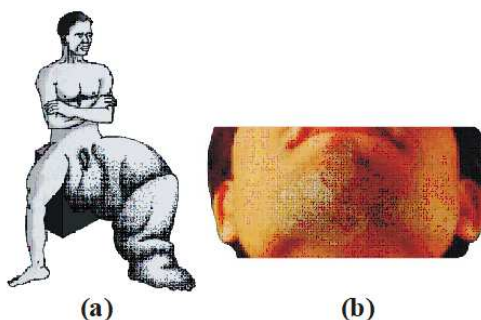
- (a) (A) is anamnestic response
 - (b) Memory cells are responsible for (A)
 - (c) Memory cells are responsible for (B)
 - (d) IgM is responsible for (A)
- (1) One (2) Two (3) Three (4) Four

145. Study the given diagram and name the labelled A, B, C and D.

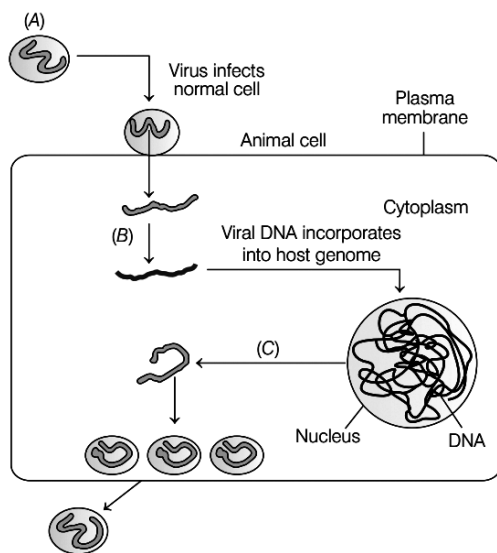


- (1) A–Sporozoites in spleen, B–Sexual reproduction of Plasmodium in RBC, C–Gametocytes in RBC, D–Fertilisation of gametocytes in mosquito's intestine
- (2) A–Sporozoites in liver, B–Asexual reproduction of Plasmodium in RBC, C–gametocytes in RBC, D–Fertilisation of gametes in mosquito's intestine (gut)
- (3) A–Sporozoites in liver, B–Asexual reproduction of Plasmodium in RBC, C–Gametocytes in RBC, D–Fertilisation of gametocytes in mosquito's salivary glands
- (4) A–Sporozoites in kidney, B–Sexual reproduction of Plasmodium in RBC, C–Gametocytes in RBC, D–Fertilisation of gametocytes in mosquito's intestine

146. The person shown in the pictures are affected by certain diseases. Identify the **incorrect** option w.r.t these diseases :



- (1) a-Caused by *Wuchereria bancrofti* and *W. Malayi*,
b-Caused by *Epidermophyton* and *Microsporum*
- (2) a-Elephantiasis, b-Ringworm
- (3) a-Caused by nematode, b-Caused by fungi
- (4) a-Acquired from soil, b-Transmitted by mosquito
147. In the given flow chart, the replication of retrovirus in a host cell is shown. Observe it carefully and fill up the blank A, B and C.



- (1) A-Bacteriophage, B-Viral DNA is produced, C-New viral RNA is produced
- (2) A-Transcriptase, B-Bacterial RNA is produced, C-New viral DNA is produced by the infected cell
- (3) A-Bacteriophage, B-Viral DNA is produced, C-New viral RNA is produced by the infected cell
- (4) A-Retrovirus, B-Viral DNA is produced by reverse transcriptase, C-New viral RNA is produced by the infected cell

148. According to world health organisation, in the given list How many are included in the definition reproductive of health ?

Physical, emotional, social, behavioural

- (1) 4 (2) 3
- (3) 2 (4) 1

149. Which contraceptive is developed by CDRI-Lucknow ?

- (1) Today
- (2) Nirodh
- (3) Saheli
- (4) Mala-D

150. In the given list how many are the properties of an ideal contraceptive method ?
- user friendly, easily available, effective, reversible, no or least side effects.

- (1) 4
- (2) 3
- (3) 2
- (4) 5

Topic : HUMAN REPRODUCTION, REPRODUCTIVE HEALTH, ORIGIN AND EVOLUTION, HUMAN HEALTH AND DISEASES, SEXUAL REPRODUCTION IN FLOWERING PLANTS, REPRODUCTION IN ORGANISM

SECTION-A

Attempt All 35 questions

151. Parturition is induced by a complex neuroendocrine mechanism involving which of the following hormones ?

- (1) Oxytocin (2) Cortisol
(3) Estrogen (4) All of the above

152. **Statement I :-** Our laws permit legal adoption and it is as yet, one of the best methods for couples looking for parenthood.

Statement II :- The reasons for infertility could be many-physical, congenital, diseases, drugs, immunological or even psychological.

Choose the correct answer from the given options :-

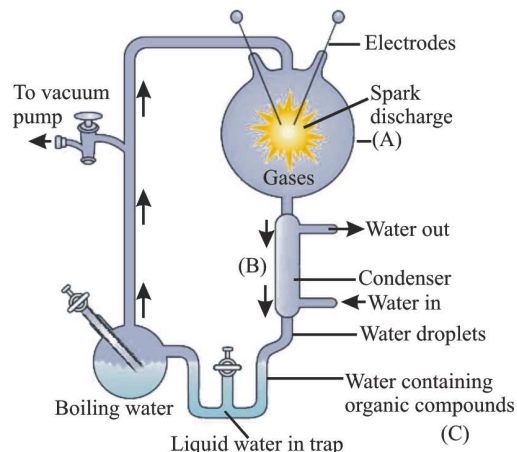
- (1) Both Statement I and Statement II are incorrect
(2) Both Statement I and Statement II are correct
(3) Statement I is correct but Statement II is not correct
(4) Statement I is incorrect but statement II is correct.

153. **Assertion (A) :** In GIFT ovum collected from a donor is transferred into the fallopian tube of another female.

Reason (R) : This female can not produce ovum but can provide suitable environment for fertilisation and further development.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

154. In following experiment of Stanley Miller A,B,C are simulation of



	(A)	(B)	(C)
(1)	Raining	Primitive atmosphere	Hot dilute soup
(2)	Hot dilute soup	Raining	Primitive atmosphere
(3)	Primitive atmosphere	Raining	Hot dilute soup
(4)	Raining	Hot dilute soup	Primitive atmosphere

155. Which statement is not true for placenta ?

- (1) It is connected to the foetus through umbilical cord
(2) It contains 100 percent maternal tissue.
(3) It provides nutrition to the foetus.
(4) It helps in removal of the waste from foetus.

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156. Which of the following is the part of amnion ?
 (1) Extra embryonic endoderm
 (2) Parietal extra embryonic mesoderm
 (3) Visceral extra embryonic mesoderm
 (4) Both parietal and visceral extra embryonic mesoderm
157. **Statement I :-** The origin of life is considered a unique event in the history of universe.
Statement II :- The gases condensed under gravitation and formed galaxies.
 Choose the correct answer from the given options :-
 (1) Both Statement I and Statement II are incorrect
 (2) Both Statement I and Statement II are correct
 (3) Statement I is correct but Statement II is not correct
 (4) Statement I is incorrect but Statement II is correct
158. **Assertion (A) :** The world we see, inanimate and animate, is only the success stories of evolution.
Reason (R) : Evolution By natural selection, in a true sense would have started when cellular forms of life with differences in metabolic capability originated on earth.
 (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.
159. Which of the following fell to form coal deposits ?
 (1) Cycads (2) Gnetales
 (3) Conifers (4) Giant ferns
160. Which of the following was about 20 feet in height and had huge fearsome dagger like teeth ?
 (1) Tyrannosaurus (2) Brachiosaurus
 (3) Stegosaurus (4) Triceratops
161. Find the total number of organisms evolved from Psilophyton in the following :
Bryophytes, Herbaceous lycopods, Sphenopsids, Ferns, Ginkgos, Conifers, Gnetales, Arborescent lycopods
 (1) 2 (2) 3 (3) 5 (4) 6
162. Which of the following statements are correct regarding origin and evolution of man ?
 (i) About 15 mya, primates called *Dryopithecus* and *Ramapithecus* were existing.
 (ii) *Homo erectus* probably lived about 1.5 mya which had a large brain around between 650-800cc.
 (iii) Pre-historic cave art developed about 18,000 years ago.
 (iv) Agriculture came around 10,000 years back and human settlements started.
 (1) (i), (iii) and (iv) (2) (i), (ii) and (iv)
 (3) (ii), (iii) and (iv) (4) All are correct
163. **Statement-1 :** *Dryopithecus* was more man like.
Statement-2 : *Ramapithecus* was more ape like.
 (1) Both statements are correct
 (2) Statement-1 is correct, Statement-2 is incorrect
 (3) Statement-1 is incorrect, Statement-2 is correct
 (4) Both Statements are incorrect

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164. Assertion : Evolution is not a directed process in the sense of determinism.

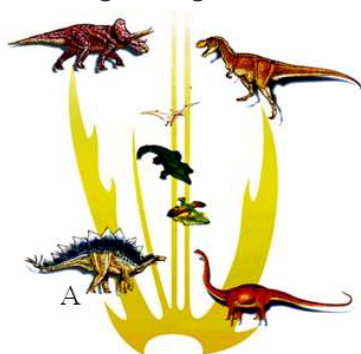
Reason : It is not a stochastic process based on chance events in nature and chance mutation in the organisms.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

165. Choose the incorrect statement.

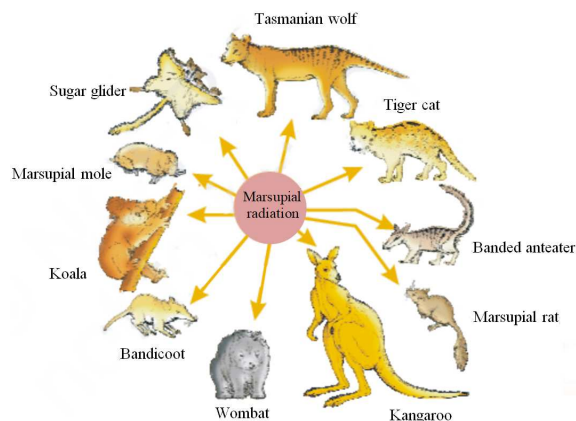
- (1) Mutations are random and directional which cause evolution.
- (2) Evolution for Darwin was gradual based on small variations.
- (3) Single step large mutation is called saltation by Hugo de Vries.
- (4) Mendel had talked of inheritable 'factors' influencing phenotype but Darwin either ignored these observations or kept silence.

166. Identify the animal-A in the family tree of dinosaurs in the given figure ?



- (1) Triceratops
- (2) Tyrannosaurs
- (3) Stegosaurus
- (4) Pteranodon

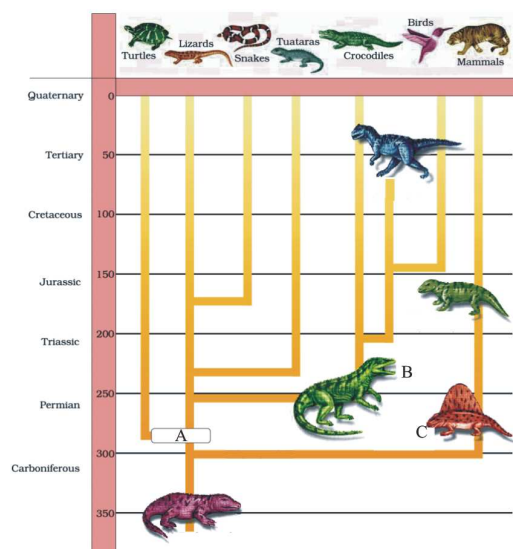
167. How many statements are correct about the diagram given below ?



- (a) Australian marsupials
- (b) Shows Homology
- (c) Example of divergent evolution
- (d) All are examples of adaptive radiation

(1) one (2) two (3) three (4) four

168.



Identify A,B and C from the above figure :-

- (1) A = Synapsids; B = Therapsids; C = Pelycosaurids
- (2) A = Synapsids; B = Thecodonts; C = Therapsids
- (3) A = Synapsids; B = Therapsids; C = Thecodonts
- (4) A = Sauropsids; B = Thecodonts; C = Pelycosaurids

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169. Suppression of South America fauna (like hippopotamus, bear, rabbit etc.) by North American fauna can be best explained by :-

- (1) Atavism
- (2) biogeographical evidences
- (3) embryological evidences
- (4) Palaentological

170. Consider the four statements (a-d) about the phenomenon of industrial melanism in England and select the option which includes incorrect statements only :

- (a) Before industrialization white winged moths were more than dark winged moths
- (b) After industrialization dark winged moths were more than white winged moths
- (c) Predators can't spot a moth against a contrasting background
- (d) Dark winged moths survived due to thick growth of white coloured lichen on trees

Options:

- (1) Statements (a) and (b) (2) Statements (a) and (c)
- (3) Statements (b) and (c) (4) Statements (c) and (d)

171. Identify the correct match from following?

	Column-A	Column-B
(1)	Invertebrate	300 MYA
(2)	Jaw less fish	350 MYA
(3)	Sea weeds	500 MYA
(4)	Coelacanth	320 MYA

172. Intensive breeding programme has created breeds that differ from other breeds for eg. Breeds of dogs but still are of same group. It is argued that if within hundred of years, man could create new breeds, could not nature have done same over millions of years? This is example of :-

- (1) Natural selection (2) Mutational breeding
- (3) Artificial selection (4) Inbreeding

173. How many of the following set of mammals are showing convergent evolution :

Mouse – Marsupial mouse, Numbat – Spotted cuscus, Mole – Marsupial mole, Lemur – Anteater, Flying squirrel – Flying phalanger, Bob cat – Koala

- (1) 5 (2) 3 (3) 2 (4) 4

174. How many of the following are homologous structure in plants ?

- (a) Potato and Ginger
- (b) Radish and carrot
- (c) Testes in male & ovaries in female
- (d) Sting of bee and scorpion
- (e) Thorn of bougainvillea and tendril of cucurbita
- (f) Sweet potato and potato

- (1) Four (2) Three (3) Two (4) Five

175. Based on observations made during a sea voyage in a sail ship called H.M.S. Beagle round the world, Darwin concluded that :-

- (i) Existing living forms share similarities to varying degree not only among themselves but also with life forms that existed million of years ago.
 - (ii) Many such life forms do not exist any more.
 - (iii) There had been extinctions of different life forms in the years gone by just as new forms of life arose at different periods of history of earth.
 - (iv) There has been gradual evolution of life forms.
- How many of the above statements are correct ?

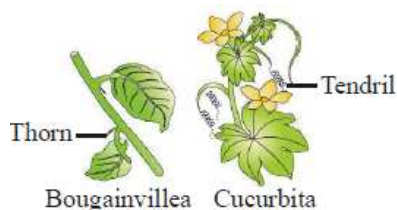
- (1) 4 (2) 3 (3) 2 (4) 1

176. When two species of different genealogy come to resemble each other as result of adaptation, the phenomenon is termed as

- (1) convergent evolution (2) divergent evolution
- (3) microevolution (4) co-evolution

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177. Which type of evolution is shown by given structures?



- (1) Parallel evolution
 (2) Convergent evolution
 (3) Divergent evolution
 (4) Retrogressive evolution
178. Match the following and give the answer for correct match :-

(A) ZIFT	(i) More than 8 blastomere stage transfer into the uterus
(B) IUT	(ii) Sperm is directly injected into the cytoplasm of ovum.
(C) GIFT	(iii) Zygote with upto 8 blastomere could be transferred into the fallopian tube.
(D) ICSI	(iv) Transfer of an ovum collected from ovary into the fallopian tube

- (1) A-i B-ii C-iii D-iv
 (2) A-iv B-iii C-ii D-i
 (3) A-iv B-iii C-i D-ii
 (4) A-iii B-i C-iv D-ii
179. In Citrus and Mango varieties ___(A)___ cells of maternal sporophyte of ovule start dividing and develop into the embryos.
 Find the "A" :
 (1) Integumentary cells (2) Nucellar cells
 (3) MMC (4) Embryosac

180. Match the following column.

Column I		Column II	
A	Parthenocarpy	1	Grasses
B	Apomixis	2	Citrus
C	Polyembryony	3	Banana

- (1) A-1, B-2, C-3
 (2) A-2, B-1, C-3
 (3) A-3, B-2, C-1
 (4) A-3, B-1, C-2
181. Read the following five statements from (A → E):-
 A. Majority of flowering plants produce hermaphrodite flowers
 B. In sea grasses the female flower reach the surface of water by the long stalk & the male flowers or pollengrains are released on the surface of water.
 C. Water pollinated plants *Vallisneria* & *Hydrilla* grows in fresh water
 D. Wind pollination is rare in grasses
 E. Chasmogamous flowers produce assured seed set even in absence of pollinators
 How many of above statements are wrong :-

- (1) Four (2) Three
 (3) Two (4) Five
182. In Majority of flowering plants functional megaspore produces :
 (1) Egg cell
 (2) One egg & three antipodal
 (3) One egg & two synergids
 (4) Female Gametophyte

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183. Assertion: Most zygote divide only after certain amount of endosperm is formed.

Reason: This is an adaptation to provide assured nutrition to the developing embryo.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

184. Select the statement that are true for sexual reproduction -

- I. Progenies produced are exact copies of their parents.
- II. Fusion of gametes occurs.
- III. Involves participation of one organism (= parent) only.
- IV. Offsprings show variations.

Options:-

- (1) I and III (2) I and IV
- (3) II and IV (4) II and III

185. In vegetative propagation of Potato and Agave, new plants arise respectively from _____ & _____

- (1) Axillary bud, Bulbil
- (2) Adventitious bud, Axillary bud
- (3) Axillary bud, Floral bud
- (4) Bulbil, Axillary bud

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. Which of the following is not included in the test tube baby programme ?

- (1) GIFT (2) ZIFT (3) IUT (4) IVF

187. Select the correct matching pair about the embryonic development of human :-

- (1) End of first month - Most of the major organ systems developed
- (2) End of second month - Eye-lids separated
- (3) End of three month - Heart formation
- (4) During fifth month - Appearance of hair on the head

188. Key concepts of Darwinian Theory of evolution are ?

- (1) Branching descent only
- (2) Natural selection only
- (3) Saltation
- (4) Branching descent and natural selection.

189. Thecodonts were evolved from ?

- (1) Synapsids (2) Therapsids
- (3) Tuataras (4) Sauropsids

190. Which one is not the theory of origin of life ?

- (1) Special creation
- (2) Spontaneous generation
- (3) Stanley Miller
- (4) Cosmic panspermia

191. Which of the following set do not represent convergent evolution ?

- (1) Wings of sparrow and wings of bat
- (2) Hind limb of rat and human
- (3) Blood proteins of man and apes
- (4) 1,2 and 3 all

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192. In __A__ a fish caught in __B__ happened to be a Coelacanth. Choose correct option for A and B

	A	B
(1)	1938	South America
(2)	1938	South Africa
(3)	1983	South Africa
(4)	1983	South America

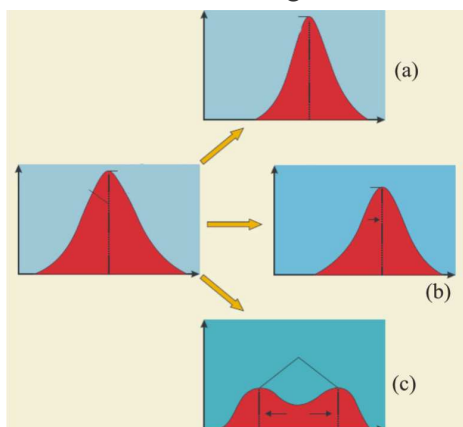
193. In the Stanley Miller experiment identify the correct combination of gases & temperature ?

	Gases	Temperature
(1)	CH ₄ , NH ₃ , CO ₂	800°C
(2)	NH ₃ , HCN, H ₂ O vapour, H ₂	1800°C
(3)	CH ₄ , H ₂ , NH ₃ , H ₂ O vapour	800°C
(4)	CH ₄ , HCN, CO ₂	8000°C

194. 1st human like being was-

- (1) *Homo habilis* (2) *Homo sapiens*
(3) *Homo erectus* (4) *Dryopithecus*

195. Recognise the figures (a), (b) and (c) given below and select the correct matching about them :-



- (1) b-directional, a-disruptive, c-stabilising
(2) b-directional, c-disruptive, a-stabilising
(3) c-directional, b-disruptive, a-stabilising
(4) a-directional, c-disruptive, b-stabilising

196. (A) Gene migration or gene flow
(B) Genetic drift
(C) Mutation
(D) Gene recombination
(E) Natural selection

Choose the option which includes factors which are known to affect Hardy-Weinberg equilibrium.

- (1) Only A, C and D (2) Only C and D
(3) All except E (4) All A,B,C,D and E

197. Genetic drift operates in -

- (1) Small isolated population
(2) Large isolated population
(3) Non-reproductive population
(4) Slow reproductive population

198. Formation of gametophyte from sporophyte without meiosis is called :

- (1) Apogamy (2) Apospory
(3) Parthenogenesis (4) Adventive embryony

199. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (1) Poaceae ; Rosaceae (2) Poaceae ; Leguminosae
(3) Rosaceae ; Solanaceae (4) Poaceae ; Solanaceae

200. **Assertion(A):** Offspring produced by asexual reproduction are not only identical to one another but also exact copies of their parents.

Reason (R): Asexual reproduction involves the fusion of gametes produced in a single parent.

- (1) A and R are true and R is the correct explanation of A.
(2) A and R are true and R is not the correct explanation of A.
(3) A is true, R is false.
(4) A is false, R is true

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

ALLEN CAREER INSTITUTE Pvt. Ltd.**Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005****Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.ac.in | Website : www.allen.ac.in**